NIFTY50 Quant Research Report

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1. Project Overview

This project implements an institutional-style quantitative research framework for the NIFTY50 index. It performs multi-strategy backtesting on India's top stocks using four trading strategies: EMA Crossover, RSI, MACD, and Bollinger Bands. The framework includes benchmark comparison, risk analysis, and portfolio analytics.

2. Data Overview

Benchmark: ^NSEI (NIFTY50)
Date Range: 2000-01-01 to 2026-05-22
Stocks Analyzed: 3 (RELIANCE.NS, TCS.NS, INFY.NS)
Data Source: Yahoo Finance (yfinance)

3. Strategy Descriptions

EMA Crossover:

Buys when fast EMA(20) crosses above slow EMA(50), sells on cross below.

RSI S

Buys when RSI(14) < 30 (oversold), sells when RSI > 70 (overbought).

MAC

Buys when MACD (12-26 EMA diff) > 0, sells when MACD < 0.

Bolli

Buys when price < lower band (SMA-2*std), sells when price > upper band (SMA+2*std).

4. Backtest Results Summary

Stock	Strategy	Equity($)	Return%	CAGR%	Sharpe	MaxDD%	WinRate%	Profit
RELIANCE	EMA	$417,816	317.82%	3.81%	0.0651	-80.73%	26.92%	2.2152
RELIANCE	RSI	$782,617	682.62%	5.53%	0.3199	-57.13%	70.59%	2.7564
RELIANCE	MACD	$894,573	794.57%	5.9%	0.0987	-77.47%	35.04%	2.3337
RELIANCE	BOLLINGER	$478,112	378.11%	4.18%	0.2423	-59.79%	67.12%	2.5664
TCS	EMA	$627,805	527.8%	5.48%	0.2269	-52.28%	35.09%	2.6333
TCS	RSI	$1,114,868	1014.87%	7.25%	0.2014	-50.38%	83.82%	4.5045
TCS	MACD	$522,848	422.85%	4.92%	0.3047	-51.79%	36.45%	2.1928
TCS	BOLLINGER	$1,210,820	1110.82%	7.51%	0.182	-58.36%	81.54%	4.135
INFY	EMA	$475,807	375.81%	4.16%	0.2364	-63.89%	32.76%	2.0979
INFY	RSI	$118,235	18.23%	0.44%	0.0244	-64.21%	66.67%	1.4057
INFY	MACD	$833,029	733.03%	5.7%	0.3224	-66.51%	39.0%	2.0375
INFY	BOLLINGER	$171,059	71.06%	1.41%	0.0771	-68.94%	63.64%	1.6265

5. Performance Analysis

Best Overall Return: TCS with BOLLINGER strategy (1110.82% return, CAGR 7.51%)
Best CAGR: TCS with BOLLINGER (7.51%)
Best Risk-Adjusted (Sharpe): INFY with MACD (Sharpe: 0.3224)
Worst CAGR: INFY with RSI (0.44%)

Strategy Averages Across Stocks:

Strategy	Avg Return%	Avg CAGR%	Avg Sharpe	Avg MaxDD%	Avg WinRate%
EMA	407.14%	4.48%	0.1761	-65.63%	31.59%
RSI	571.91%	4.41%	0.1819	-57.24%	73.69%
MACD	650.15%	5.51%	0.2419	-65.26%	36.83%
BOLLINGER	520.00%	4.37%	0.1671	-62.36%	70.77%

6. Conclusions

1. RSI Strategy shows the highest average CAGR (5.51%) and strongest performance across TCS and RELIANCE, but struggles on INFY (only 0.44% CAGR).

2. MACD Strategy provides the most consistent risk-adjusted returns with the highest average Sharpe ratio (0.2419) and solid CAGR across all three stocks.

3. Bollinger Bands performs best on TCS (7.51% CAGR) but poorly on INFY (1.41% CAGR), indicating stock-specific suitability.

4. EMA Crossover delivers moderate returns but suffers from higher drawdowns and lower win rates (26-35%), typical of trend-following strategies during range-bound markets.

5. No single strategy consistently outperforms across all stocks - strategy selection should be stock-specific and market-regime aware.

6. Buy & Hold returns generally exceed strategy returns over the full 26-year period, highlighting the challenge of market timing vs. long-term holding in a bull market.

7. Recommendations

1. Use MACD Strategy as a baseline for multi-stock portfolios - it offers the best risk-adjusted returns across different stocks.

2. Apply RSI Strategy selectively - excellent for large-cap liquid stocks (RELIANCE, TCS) but avoid on more volatile growth stocks (INFY).

3. Consider parameter optimization - the default parameters (20/50 EMA, 14 RSI, etc.) may not be optimal. Grid search over parameter space could improve returns.

4. Combine strategies into an ensemble model - for example, use MACD for trend direction confirmation and RSI for entry timing.

5. Add risk management rules - position sizing, trailing stop-losses, and portfolio-level risk limits to reduce drawdown severity.

6. Extend analysis to all 15 NIFTY50 stocks in the config and include sector-wise performance decomposition.

7. Incorporate machine learning models (e.g., Random Forest, XGBoost) for regime detection to dynamically select the best strategy.

8. Backtest with transaction costs and slippage models more realistic than the current 0.2% flat commission.